

EUROPEAN ECONOMICS ANALYST

Spillovers from the Global AI Capex Cycle

- The AI investment cycle continues unabated, with global AI investment exceeding \$1tn this year. While most of this spending is concentrated in the US and Asia, the global investment boom is showing signs of positive growth spillovers into Europe, partly offsetting headwinds from energy prices. In this *Analyst*, we review the evidence and estimate the likely contribution of the global AI capex cycle to European GDP growth.
- To do so, we leverage three different approaches. First, we allocate global AI-related spending across countries and sectors by mapping the geographical distribution of new data centre capacity additions, assuming a spending mix similar to that of the US. We feed the resulting country-sector spending changes into a global input-output model to compute the direct and indirect effects of global AI spending growth on European GDP growth. This approach suggests that global AI capex will add 0.10pp-0.15pp to European growth in 2026.
- Second, we adopt a bottom-up approach that computes the domestic value added of the European data centre build-out and of AI-related exports, which also points to an estimated AI investment contribution to GDP growth of around 0.1pp in 2026. Third, we examine granular industrial production data in AI investment-exposed sectors and come up with a similarly sized impulse to growth.
- We cross-check our estimates using changes in earnings expectations for European AI-exposed companies. We find that 2026 earnings growth expectations since the onset of the AI cycle have risen by around EUR 12bn, equivalent 0.05% of European GDP.
- Taken together, our exercise points to a European growth boost of 0.05pp to 0.15pp, with a mean estimate of 0.10pp. Two groups of countries are set to benefit most from the capex cycle. First, countries that attract physical investment in data centres, such as Scandinavian economies and Iberia, are benefitting from domestic construction and fitting activity. Second, manufacturing economies integrated in the global semiconductor and electrical equipment supply chain benefit from incremental demand globally, including Germany.
- Looking ahead, projections by our global economics team suggest that global AI-related investment will continue at a firm pace, implying that the positive

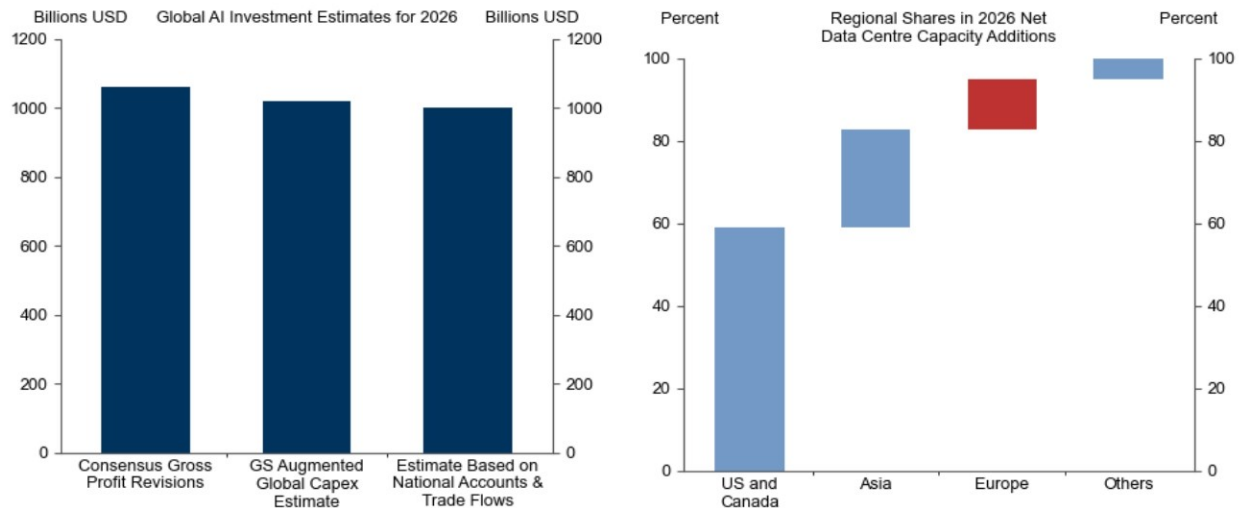
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impulse to European growth observed in 2026 should continue into 2027. The uncertainty around our estimates is two-sided. On the one hand, investment designed to expand production capacity in affected sectors and positive sentiment effects could understate the contribution of the global AI buildout to European activity. On the other hand, mismeasurement and crowding out of non-AI activity could imply a less positive contribution to European growth.

Spillovers from the Global AI Capex Cycle

The AI investment cycle continues unabated, with our latest estimates pointing to global AI investment exceeding \$1tn this year (Exhibit 1, left). Most of the hardware investment is concentrated in the US and Asia, that together account for over 80% of the global net data centre additions. Europe only represents around 10% of global additions to data centres this year (Exhibit 1, right).¹

Exhibit 1: Global AI Investment Reaching \$1tn in 2026

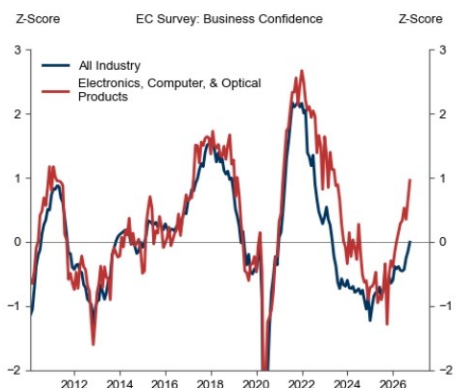


Source: Goldman Sachs Global Investment Research, Bloomberg, Haver Analytics, 451 Research

That said, recent commentary from European central banks and business survey data point to signs of positive spillovers of the global investment cycle into Europe (Exhibit 2), which could explain the surprising resilience of the activity dataflow to the energy shock observed so far.

Exhibit 2: Some Signs that Europe Is Benefitting from AI-Related Demand

Geography	Date	Context	Quote
Euro Area	10-Sep	ECB's Monetary Policy Statement	Increased AI-related activity is visible in digital services, business investment and exports.
	14-Sep	Speech by ECB President Lagarde	AI-related borrowing accounted for about a quarter of the growth in credit to firms in the first quarter of this year. Since euro area long-term rates move with US yields, Europe will bear part of the price of this boom in its own borrowing costs.
Sweden	24-Sep	Riksbank's Monetary Policy Report	Manufacture of computers and electronics has also experienced strong growth and value added is now around 50 per cent larger than in 2019. This industry may benefit from the expansion of AI and data services, but also from military rearmament. The upturn in investment can be related to major investments in defence and AI technology, which have also been highlighted by the companies responding to the Riksbank's most recent Business Survey.
Norway	24-Sep	Norges Bank's Monetary Policy Report	Data centre buildout, technology investment and capacity expansion in tourism are boosting investment.
Euro Area	23-Sep	Flash PMIs	Manufacturing, spearheaded by Germany, is enjoying its best growth spell for over four years, spurred by rising AI and defense spending
UK	23-Sep	Flash PMIs	AI investment and greater defence spending were highlighted as factors supporting manufacturing production, while weak consumer demand was reported as a growth headwind.



Source: S&P Global Market Intelligence, ECB, Riksbank, Norges Bank, European Commission, Goldman Sachs Global Investment Research

The European economy can benefit from the global AI investment cycle through two

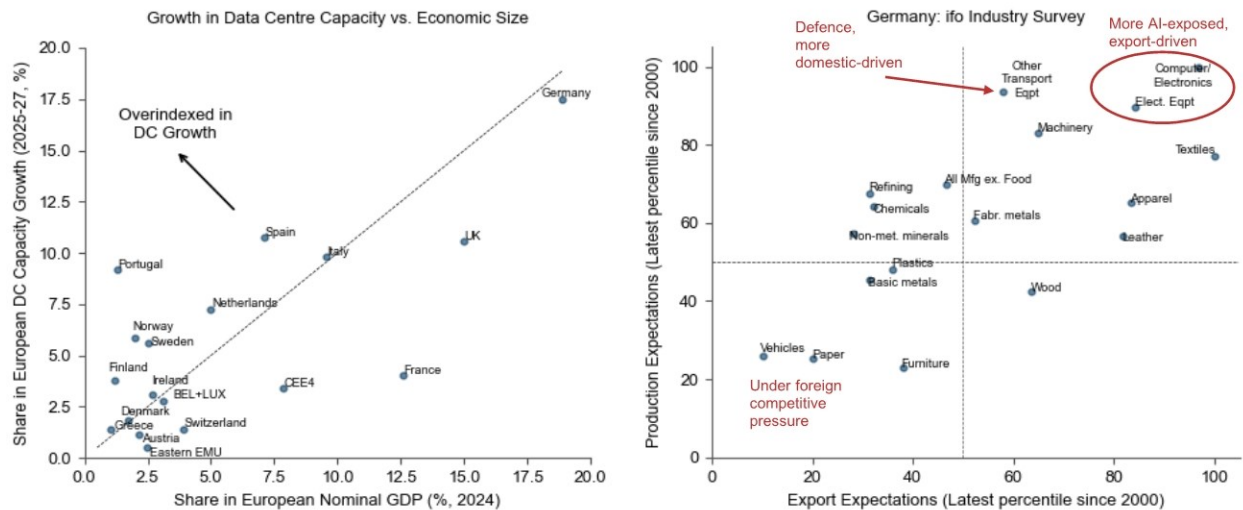
¹ In this note, "Europe" refers to the EU27, the UK, Norway, and Switzerland.

main channels.

First, while less prominent than elsewhere, local data centre build out should have a modest positive effect on European growth via increased activity in construction and fitting, even if the computing hardware is largely imported ([Exhibit 3, left](#)).

Second, Europe should benefit from increased demand for AI-exposed goods, including semiconductor-related capital goods, electronics, and electrical equipment. Indeed, business surveys in these sectors point to export and production expectations at or close to historical highs ([Exhibit 3, right](#)).

Exhibit 3: Two Possible Channels: Domestic Data Centre Construction and Spillover from Investment Abroad



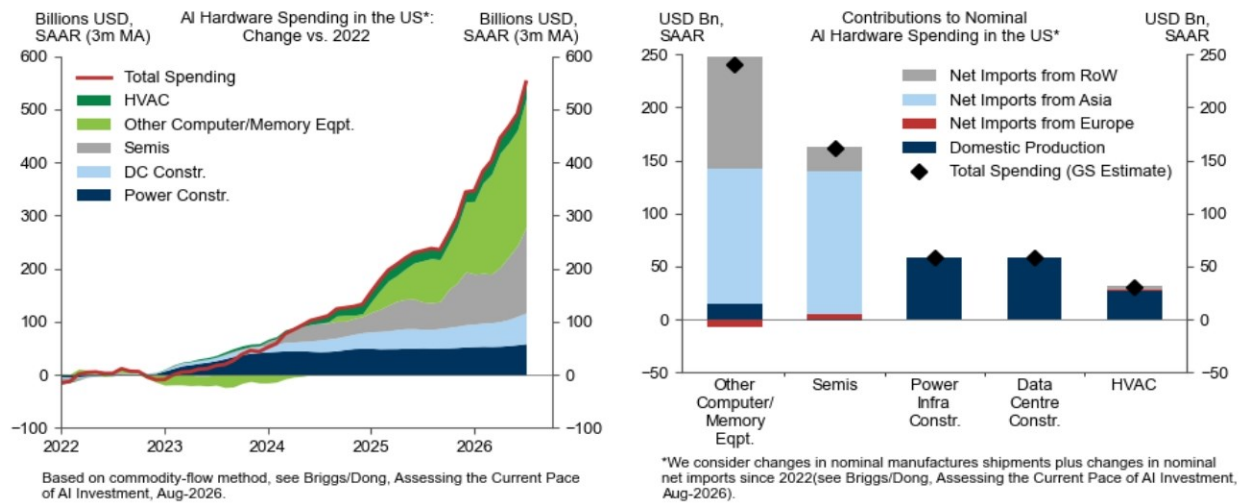
Source: European Data Centre Association, ifo, Haver Analytics, Goldman Sachs Global Investment Research

In this *Analyst*, we thus review the evidence of these two channels and employ three different approaches to estimate the likely contribution of the global AI capex cycle to European GDP growth.

Modelling Spillovers with Input-Output Table

First, we adopt a model-driven approach that leverages input-output tables to capture direct effects and spillovers from the global AI cycle.

To do so, we leverage detailed metrics on the composition of US AI hardware spending compiled by our Global team colleagues by replicating the US national accounts commodity-flow method of estimating investment. Their approach enables us to break down US AI spending by component and to track how much of that is resulting from higher net imports as opposed to increased domestic production.

Exhibit 4: US Data Suggests Most AI-Related Investment is on Imported Semis and Computer Equipment

Source: US Bureau of Economic Analysis (BEA), Haver Analytics, Goldman Sachs Global Investment Research

Given the lack of comparable timely and granular data in other jurisdictions, we assume regional AI spending is proportional to net computing power adds across regions, using data compiled by our equity research colleagues.² Furthermore, we assume that the spending mix outside of the US broadly mirrors that observed in the US with respect to both components and geographical sourcing.³

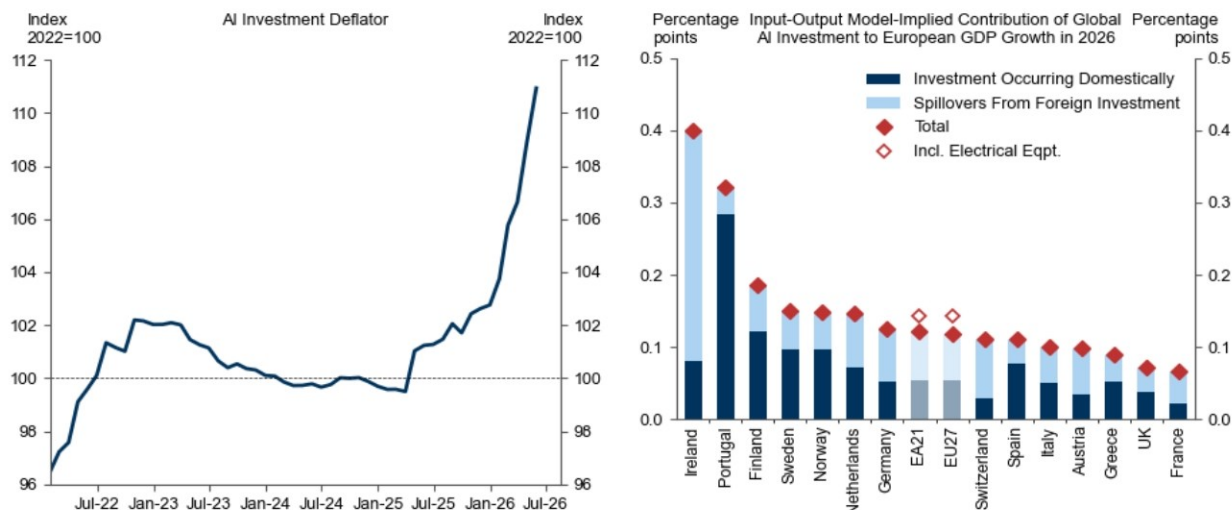
These assumptions allow us to decompose global AI investment growth into changes in spending across countries and industries. After adjusting nominal demand growth for the rapid inflation in input costs observed this year (Exhibit 5, left), we are able to feed a real demand growth vector to input-output tables⁴ that enables to track how an increase for final-demand products reverberates up through the supply chains to producers of intermediate goods (Exhibit 5, right).

² To allocate spending within Europe, we leverage data on expected additions compiled by the EU data centre association (EUDCA).

³ When extrapolating the US data to global spending, we adjust our assumptions on the likely origin of products to avoid overstating the role of countries like Mexico that benefit from their proximity to the US.

⁴ We use the 2022 OCED industry-by-industry input-output table as they cover Taiwan.

Exhibit 5: After Adjusting for Cost Inflation, Our Input-Output Based Model Points to an AI Investment-Related Boost to Euro Area GDP Growth of Around 0.10pp-0.15pp



Source: US Bureau of Economic Analysis (BEA), Haver Analytics, OECD, Goldman Sachs Global Investment Research

Our model suggests that global AI investment is contributing around 0.10pp to European growth, with just over half of this impulse arising from spillovers of investment occurring outside the region. This estimate rises to 0.15pp when adopting a broader definition of AI spending to include specialized electrical components beyond heating, ventilation, and air conditioning (HVAC). If applied to East Asian economies, our approach would be consistent with previously estimated AI-related growth impulses, such as the roughly 9pp contribution estimated for Taiwan this year.

A Bean-Counting Exercise

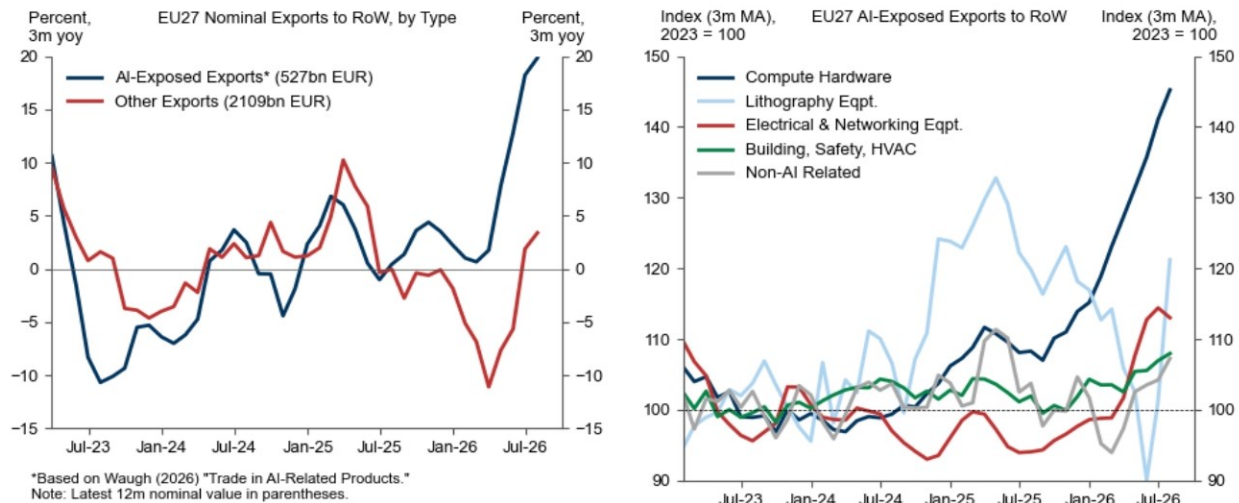
Second, we adopt a bottom-up approach that attempts to size the impulse on a component-by-component basis.

With respect to domestic spending, we again consider the increase in data centre investment in Europe using estimates from GIR equity research, and, in line with their findings, assume that (a) each MW of added data centre capacity costs around \$42 million, (b) around 70% is foreign-sourced compute equipment, with the rest accounting for the construction cost of the “shell” of the data centre that should have a substantially higher domestic GVA contribution.⁵

With respect to spillovers from foreign investment, we leverage granular trade data from Eurostat and a classification of AI-exposed products identified by the literature.⁶ We find that European Union AI-exposed goods exports to the rest of the world have clearly outperformed the rest of the local manufacturing complex, supported by computer hardware, lithography equipment, and electrical and networking equipment (Exhibit 6).

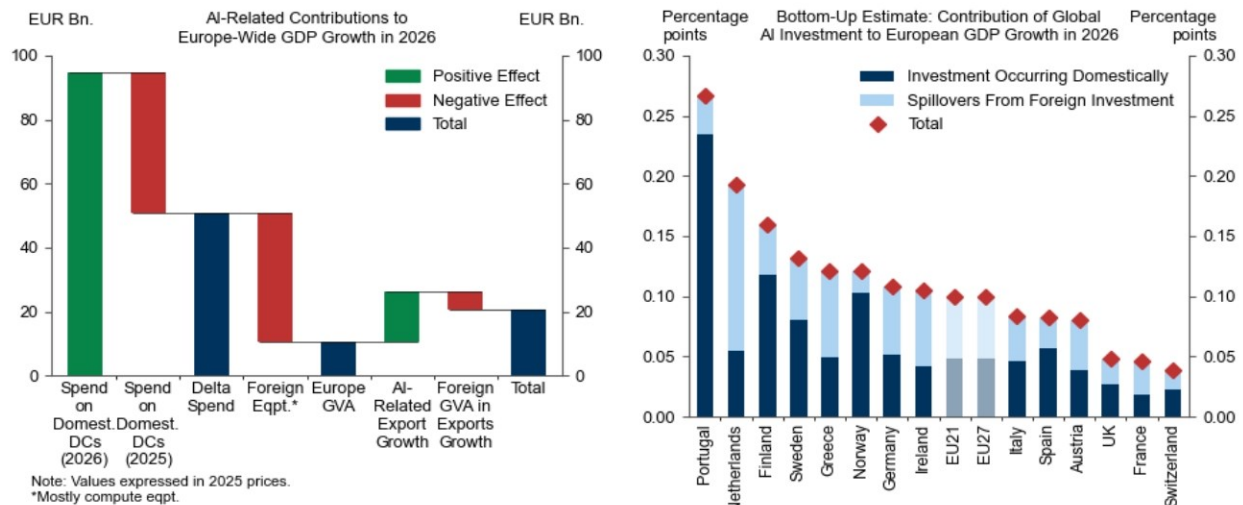
⁵ We assume that of the “shell” cost, around half is construction spending that is domestic and the rest is equipment spending allocated within Europe proportional to shares in electrical equipment manufacturing GVA.

⁶ “Trade in AI-Related Products” by M. Waugh, Federal Reserve Bank of Minneapolis and NBER (2026).

Exhibit 6: AI-Exposed Exports Have Outperformed

Source: Eurostat, Haver Analytics, Goldman Sachs Global Investment Research

After accounting again for the meaningful inflation observed in AI-exposed equipment, we estimate that AI-exposed exports are currently rising 3pp per year faster than other exports. Once accounting for the share of domestic GVA in gross exports, we estimate this outperformance added around EUR 10bn to GDP growth in 2026. Paired with our estimate of around EUR 10bn of contribution from investment occurring in Europe, we estimate a total contribution to growth of around EUR 20bn in 2025 prices, adding just under 0.10pp to Europe-wide growth, close to our model-based estimates (Exhibit 7).

Exhibit 7: Our Bottom-Up Exercise Suggests AI Investment Growth is Adding Around EUR 20bn to European GDP Growth in 2026

Changes in domestic spending (as opposed to levels) are relevant for the impulse to real GDP growth.

Source: Haver Analytics, Goldman Sachs Global Investment Research

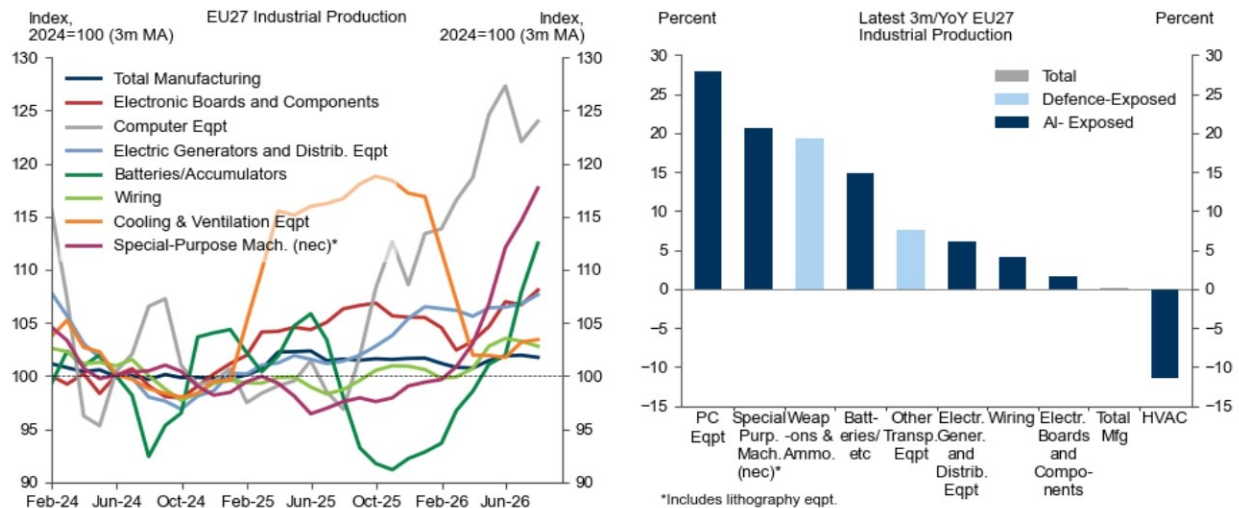
An Industrial Production-Based Approach to Assess the Impact of AI Investment

Finally, we turn to industrial production (IP) data to size the impulse of AI investment in

Europe.

We consider EU-wide IP data at the granular sub-industry level and identify seven subcategories that are particularly exposed to the AI investment cycle. Production in these areas is clearly outperforming the rest of the manufacturing sector, with growth rates comparable to those observed in areas that are benefitting from increased defence spending ([Exhibit 8](#)).

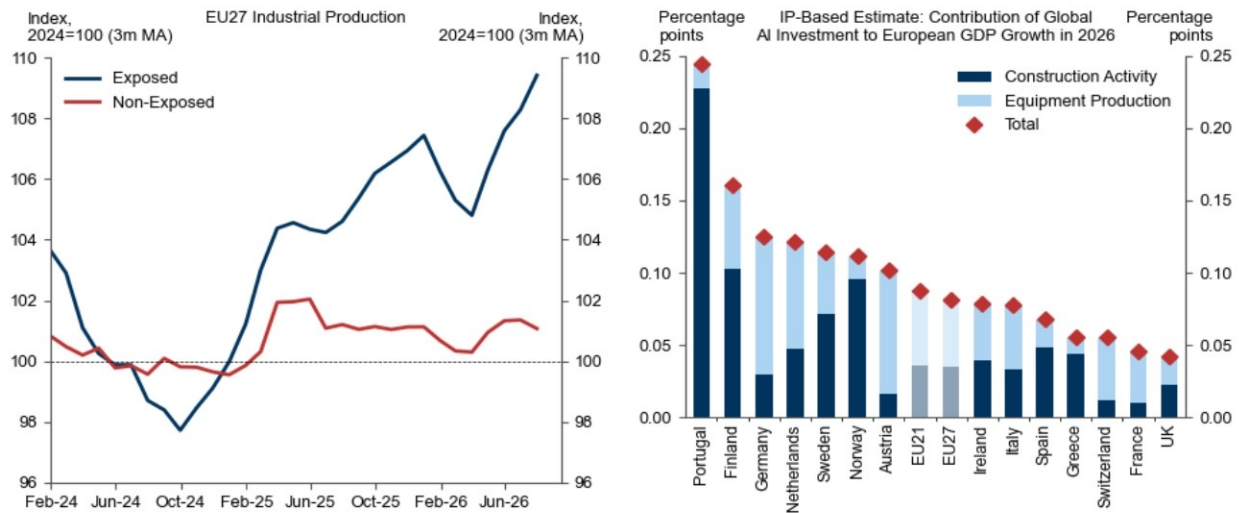
Exhibit 8: Industrial Production in Exposed Sectors is Growing Substantially Faster Than Other Categories



Source: Eurostat, Haver Analytics, Goldman Sachs Global Investment Research

Over the last 12 months, activity in AI investment-exposed sectors outperformed the rest of the industrial sector by just over 5pp. Given an approximate share in total manufacturing production of around 6% and scaling for the manufacturing share in total GDP, we estimate an approximate contribution from faster IP growth to EU-wide GDP growth of around 0.05pp ([Exhibit 9](#), left).

Given the lack of similarly granular data on construction, we are unable to separately estimate a construction impulse from monthly series and thus assume the same contribution from building activity estimated in our bean counting exercise, for a total effect of slightly less than 0.10pp on region-wide GDP growth this year ([Exhibit 9](#), right).

Exhibit 9: This Approach Also Yields a Growth Impulse Estimate of Around 0.10pp

Source: Eurostat, Haver Analytics, Goldman Sachs Global Investment Research

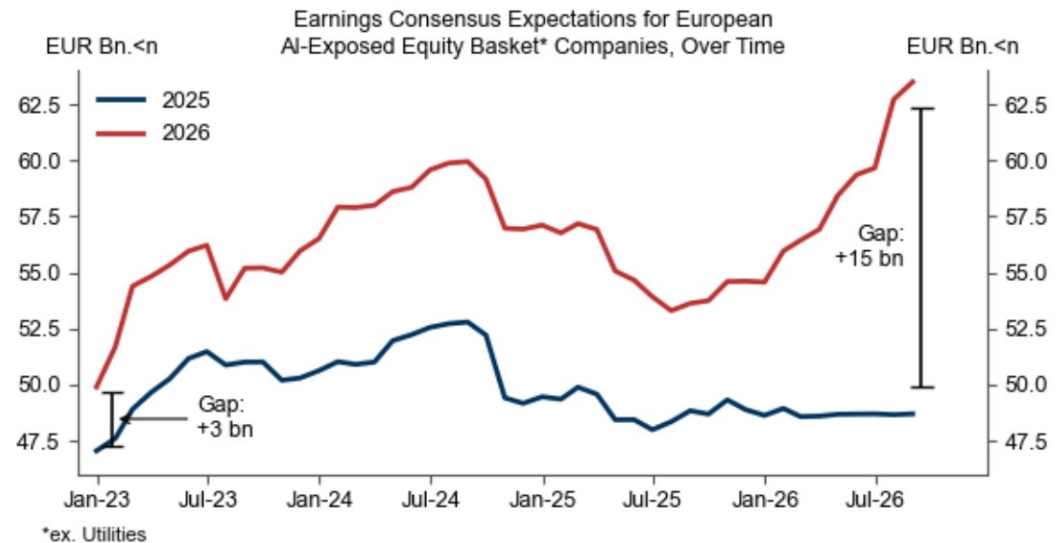
The Signal from the Stock Market

We cross-check our estimates with the signal from the stock market.

Similarly to previous work by the Global Economics team, we consider how consensus expectations for the earnings of European listed AI-exposed companies have evolved since the end of 2022, as a proxy for the contribution of the AI cycle to European corporate activity.

The expected change in earnings between 2025 and 2026 rose from just EUR 3bn at the end of 2022 to around EUR 15bn now, suggesting that the news around AI investment added around EUR 12bn to growth expectations, corresponding to around 0.05% of European GDP.

Exhibit 10: Consensus Expectations of European Listed AI-Exposed Companies Suggest the AI Cycle Added Around EUR 12bn to Earnings Growth



Source: Haver Analytics, Bloomberg, Goldman Sachs Global Investment Research, Factset

This approximate market-based estimate is slightly below those arising from previous methods. That said, it misses the contribution from non-listed companies, from European activity of foreign multinational corporates (and, similarly, does not account for operation of European companies abroad), and from most construction activity.

Bringing It All Together

We bring our estimates together in [Exhibit 11](#).

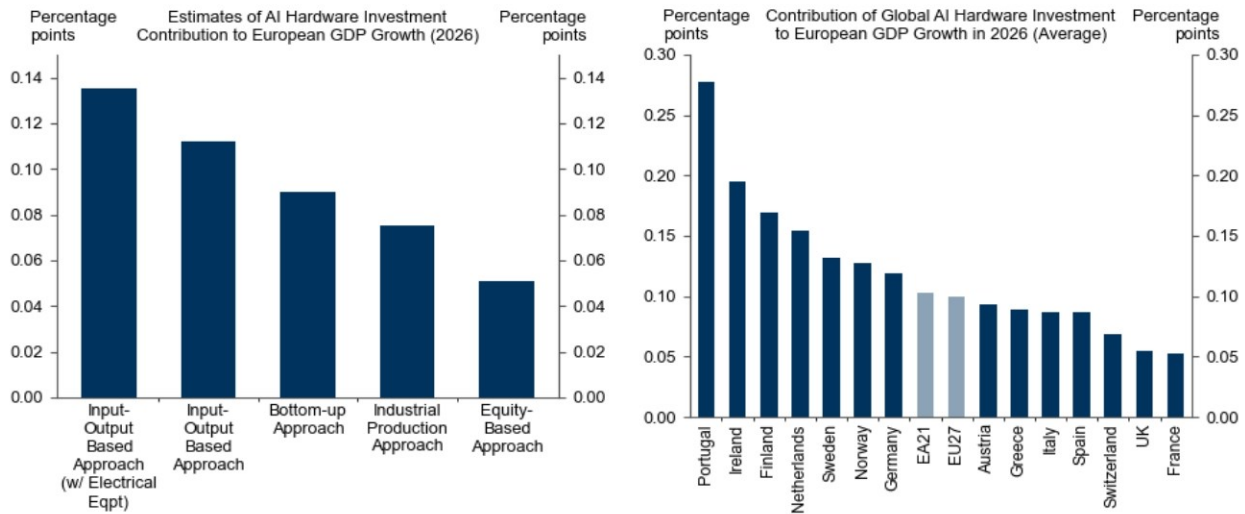
Our estimates of the impulse from the global AI investment cycle to region-wide growth range from 0.05pp to 0.15pp, averaging around 0.10pp for this year. Our input-output table-based approach that models the entire supply chain yields larger estimates, while a narrow market-based estimate sits at the lower-end of the range.

Averaging the country-level estimates from our three approaches suggests that two groups of countries are set to benefit most from the capex cycle.

First, countries that attract meaningful physical investment in data centres, such as Scandinavian economies and Iberia, are benefitting from domestic construction and fitting activity. Second, manufacturing economies integrated in the global semiconductor and electrical equipment supply chain benefit from incremental demand globally, including Germany, Ireland and the Netherlands.

Countries that do not attract a meaningful share of new data centre net additions (even though they might be accounting for a larger share of the existing stock) and are less manufacturing-oriented, such as the UK and France, screen as less positively affected than others.

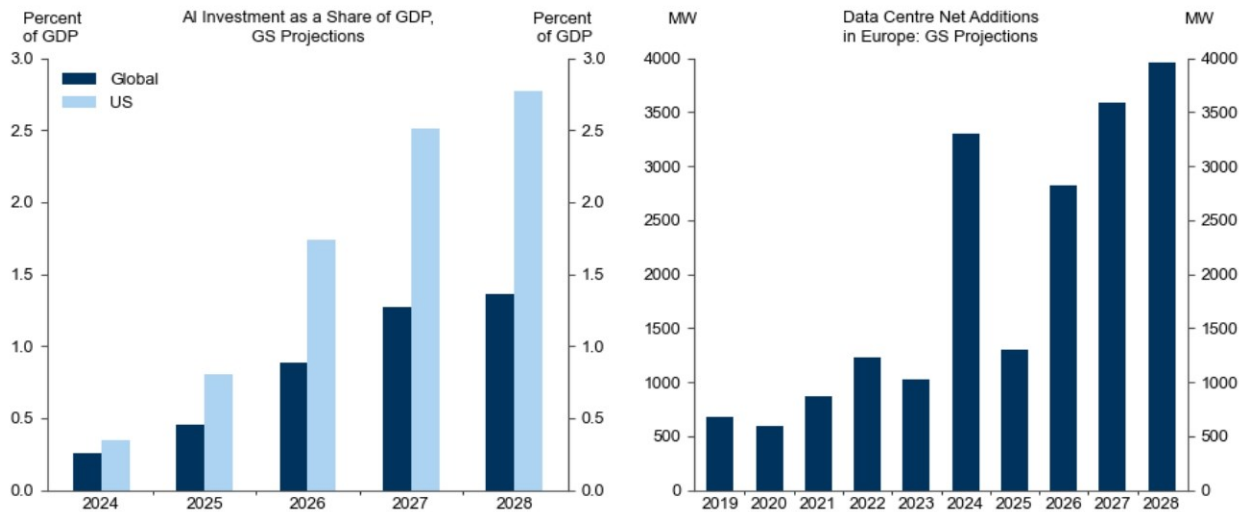
Exhibit 11: We See an AI-Related Boost of Around 0.1pp This Year, With Larger Effects in Portugal, Scandinavia, Ireland, the Netherlands, and Germany



Source: Haver Analytics, Goldman Sachs Global Investment Research

Looking ahead, projections by the Global Economics team suggest that global AI-related investment will continue at a firm pace. Domestically, investment in new data centres across the region should also continue to gain momentum in coming years. We thus expect the positive impulse to European growth observed in 2026 to continue into 2027 ([Exhibit 12](#)).

Exhibit 12: The AI Investment-Related Growth Impulse Should Remain Positive in 2027



Source: Haver Analytics, Goldman Sachs Global Investment Research, 451 Research

We see two-sided uncertainty around our estimates.

To the upside, investment designed to expand production capacity in affected sectors, such as projects announced in [Ireland](#) and [Germany](#), investment in power infrastructure, and positive sentiment effects could understate the contribution of the global AI buildout to European activity.

To the downside, it is possible that the national accounts underestimate the ongoing

investment in ICT equipment as, like in the US, statistical agencies largely employ a commodity-flow approach to measuring fixed investment with fixed capital formation ratios that could miss out the new role of ICT equipment as capital (rather than intermediate) goods. Furthermore, burgeoning AI-related activity could be crowding out other investment via its effect on technological component prices and on capital markets.

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